

ARTICLE

Privacy-Preserving Framework for Safeguarding Healthcare Data with Embedded Authentication Across Federated Cloud Platforms

S. Sivakumar^{1,*} and T. Karthikeyan²

¹Assistant Professor, PG Department of Computer Science, Government Arts College, Udumalpet, India.

²Department of Information Technology, University of Technology and Applied Sciences - Salalah, Thumrait Rd Thumrayt Street – OM 211, Sultanate of Oman.

*Corresponding Author: S. Sivakumar. Email: drsivacsphd@gmail.com

Received: 20 June 2024; Accepted: 09 September 2024; Published: 30 December 2024

ABSTRACT: Cloud Computing (CC) technology, because of its distributed environment, possesses more complicated reliability issues and privacy preservation issues of its outsourced data. The proposed design influences cloud infrastructures to give cost competitiveness, fast exploitation, scalability, and flexibility to have room for unstable workloads. The research system could be utilized to safeguard diverse medical health care records. Privacy-protecting storage framework presents a compressive sensing method in a multi-cloud design. The major motivation of this work is to introduce a new secure system for Medical Healthcare records. The proposed Secure Medical Healthcare Content Protection System (SMCPS) consists of two major steps: (i) Hybrid Key Generation (HKG) and (ii) storage management. In the initial step of the work, key values are created for medical healthcare records by using two encryption techniques, such as Advanced Encoding Standard (AES) cryptography and Message Digest (MD5) algorithm for safe multiparty computation. This proposed Hybrid Key Generation (HKG) algorithm is different from the usual key generation algorithm, since the proposed HKG algorithm adds the key values of AES and MD5. A new key value is created from these two algorithms, thus increasing the security level of the system. The HKG algorithm creates dynamic and representative keys of medical healthcare records; it is computationally resourceful to calculate and contrast, in addition, it needs a small storage. In the second step of the work, a Compressive Sensing (CS) based framework for storage management of medical healthcare records was developed. The distributed matching engine achieves greater adaptability, and it is created to work among different healthcare records gathered from the UCI machine learning vault. The applicability is further examined for the applicability of sparse coding-based frameworks, including a variety of medical healthcare records, which influences the ability of the Multi Cloud Computing (MCC) environment. The experimentation work is performed by using the Miscellaneous database for watermark pattern, and any medical healthcare records are taken as input in the MCC environment results prove that safe watermark recognition in the CS domain is possible for medical healthcare records.

KEYWORDS: Cloud Computing (CC), Secure Watermark Detection, Secure Multiparty Computation, Privacy Preserving, Compressive Sensing (CS), Cryptography Algorithms.

1 Introduction

Cloud Computing (CC) is considered to be an Information Technology (IT) structural design for ventures of future generation, as it consists of its extensive list of extraordinary benefits in IT records [1]: on-demand self-service, transfer of risk, rapid resource flexibility, location self-governing



resource pooling, present network access, and usage-based pricing [2]. By utilizing information technology, CC has become the nature of businesses with insightful entailments as a troublesome technique. One basic facet of this Model shifting has the basic facet of data centralization or outsourcing to the Cloud. From the standpoint of users, comprising individuals and IT enterprises, storage of data in distant means to the cloud provides advantages in an adaptable, on-demand way: despite the trouble in storage management, avoidance of resource expenses on hardware, and general data access along with site independence, software, and human resources maintenance, and so on [3].

In the meantime, social networks and the internet have attained an enormous growth, which makes it extremely easy for users to gather a huge volume of data on medical health care from varied sources without pointing out the copyright information of that data. Simultaneously, CC transforms these benefits more alluring than yet, as it carries novel and tough security threats well in the direction of outsourced data of users. Time to time exposing of instances of outages and security breaches occur in striking cloud services [4]. Cloud Service Provider (CSP) provides a variety of inspirations as a secondary to perform faithless laying the direction of the cloud users, which concerns their outsourced data status. By removal of data, CSP may regain storage as an instance for economic reasons which is not often used, or still conceal data loss events in order to preserve a reputation [5]. In brief, though outsourcing data to the cloud for long-term significant storage is inexpensively attractive, it does not provide any assurance instantly on data accessibility and integrity. In case this difficulty is not addressed appropriately, it might obstruct the triumph of cloud architecture.

For resolving this issue, proposed work uses the methodology called AES cryptography [6] and Message Digest (MD5) algorithm [7] which facilitates watermark pattern that owner wants to share the data devoid of dripping data of local copy and radical decreasing of communication consequently and computational overhead as contrasted for clear-cut data sharing among the users of the cloud. In addition, the cloud user might desire to gain the benefit of the storage in the cloud, and simultaneously, work with copyright owners for watermark recognition whilst maintaining the self-gathered medical healthcare data as private. The watermark pattern owners are in require of maintaining their watermark patterns as private for the particular duration of watermark recognition. An authorized cloud providing storage services as well wishes to take part in recognition of watermark initiated by the users, or kick off watermark recognition devoid of taking part in the users, for ensuring whether the uploaded medical healthcare data is copyright protected. Encrypted medical healthcare data provides an additional advantage of storing data and enables encrypted domain recognition of the watermark in the cloud, so that the encrypted data could be reprocessed if the image data holder is required to work with other watermark owners for safe watermark recognition at a later time. Additionally, the proposed work provides a Compressive Sensing (CS) based watermark recognition, which influences a safe MCC environment.

2 Literature Review

Kim et al [8] show an open evaluation convention for educational media information outsourced to the distributed storage. By making use of irregular esteems and also a homomorphic hash work, the exhibited convention ensures information protection for the cloud, notwithstanding the Third-Party Auditor (TPA). In addition, it is sheltered, contrary to losing an attack and a temper attack. Furthermore, the presented protocol is capable of aiding entirely dynamic auditing. Security and performance investigation outcomes prove that the presented method is safe whilst assuring the least additional computation costs. The foremost concern is how to guard the private data of the image whilst facilitating image computation. In order to deal with this basic confrontation, we should learn the privacy prerequisites in outsourcing Scalar Invariant Feature Transform (SIFT)

computation and need to present SecSIFT, which is a high-performance privacy-preserving SIFT feature recognition system. One of the general methods in preceding private image computation involves enciphering the private image in a public key-based homomorphic technique, which would facilitate the real processing techniques as intended for the plaintext domain to be carried out over the ciphertext domain. The system is not restricted by the competence restrictions of the homomorphic encryption method in the challenge of these works. Qin et al [9] presented a novel system that could share the computation processes of SIFT to a self-regulating group, supportive cloud servers, and it remains as the outsourced computation processes as simple as possible in order to avoid the utilization of a homomorphic encryption scheme. Thus, it enables the implementation of realistic computation with communication complexity. Comparison between SecSIFT and original SIFT on image benchmarks occurs which whilst preserving the privacy in a competent manner in this experimentation.

Voloshynovskiy et al [10] present the multimedia content privacy-preserving detection for a cloud-based Bag-of-Feature (BoF) framework. They systematically model utilization of geometric information, which could be a shared secret and drawn from a tradeoff among the ability of detection, computational load, and privacy. In addition, a recommendation of a descriptor ambiguity technique is performed that presents ambiguity to the server with the real interest of data users. Zhang et al [11] guarded the private data; the BGV encryption technique is utilized by the presented model to encipher the private data, which applies cloud servers to carry out the high-order back-propagation method competently on encrypted data for training of an in-depth computation model. In addition, the better accuracy of the presented method of the sigmoid function as a polynomial function helps the safe computation of the activation function with the encryption of BGV. The BGV encryption method includes simple encryption operations as well as the decryption operations that are carried out by the client, whilst the computation tasks are entirely carried out on the cloud. Outcomes of experimentation prove that our method is enhanced roughly by 2.5 times in the training competence, which is contrasted with the deep conventional computation model, which is devoid of exposing private data by utilizing cloud computing, comprising ten nodes. Most significantly, the presented model is greatly scalable by means of applying more cloud servers, which is mainly appropriate for big data.

Kang et al [12] presented a structure, simultaneously to examine the applicability of the previous Compressive Sensing (CS)-based on multimedia compression, and is firmly compressive multimedia “trans-detecting” techniques which rely upon effective coding to solidly appropriate compressive detection of interactive media information over a cloud-aware situation. Besides, they also inspect the utilization of past scanty coding-based structures for various sight and sound applications by utilizing the well-designed capacity of a media cloud. To think about the numerous basic confronts for multimedia cloud computing in purpose, security, and network/device heterogeneities, for instance, which examine the applications of sparse coding methods and CS in multimedia applications and delivery. Likewise, build a fused cloud-mindful structure for protection, saving mixed media applications by methods for inadequate coding. The above-stated techniques are used to simplify a single cloud computing environment, and storage management issues are resolved in some of the works only. How the data is protected by utilizing algorithms in the MCC environment is discussed in this way.

Alqahtani and Sant [13] presented solution would consider the benefits of multi-clouds, data cryptography, and data compression to protect the distributed data; by dividing the data into sections, enciphering the sections, compressing the sections, sharing out the sections by means of multi-clouds whilst maintaining one section on the mobile device memory that would secure taking out the data when the shared out segments are phished. Liu et al [14] propose a competent and several domains’ crossways Privacy-preserving Skyline Computation structure, which is known as

EPSC. Safer calculation of a skyline outcome from various service providers in the EPSC framework offers the client improved services. Meanwhile, the least privacy disclosure would be extracted from one service provider to another service provider for the extraction of privacy disclosure for the duration of skyline computation. In Particular, for controlling the privacy of service provider disclosure and approximate attainment of real-time skyline processing as well as transmission, present an Efficient Secure Vector Comparison (ESVC) protocol for building EPSC that is completely dependent upon new methods: Fast Secure Permutation Protocol (FSPP) as well as Fast Secure Integer Comparison (FSIC) protocol. Computation of skyline outcome is allowed by both the protocols synergistically in a privacy-preserving manner. A thorough security examination can attain multi-domain skyline computation as proved by the EPSC framework with devoid of disclosing it to everyone that the susceptible information. Performance assessments, in addition to widespread simulations, also show the EPSC's effectiveness in the words of giving skyline computation and transmission whilst which decreases the privacy disclosure of crosswise diverse domains. Fast Secure Integer Comparison Protocol, as well as Fast Secure Permutation Protocol, have been developed. The skyline set is calculated in a privacy-preserving and competent manner by means of widespread simulation.

The issue of the present work is that even though outsourcing healthcare data is inexpensively attractive to the multi-cloud for long-standing significant storage as it will not instantly provide any assurance on data accessibility and integrity. In case this difficulty is not appropriately addressed, this might obstruct the cloud architecture achievement. To solve this problem, the Secure Medical Healthcare Content Protection System (SMCPS) is being proposed for the large-scale medical healthcare records. The advantage of this SMCPS system is that it can be applied easily to a multi-cloud environment and is more secure for medical healthcare records. By adding the CS framework, the size of the original file is reduced, which reduces the storage space and decreases the time complexity of storage.

3 Proposed Methodology

In this work, a new Secure Medical Healthcare Content Protection System (SMCPS) is proposed for large-scale medical healthcare records. This SMCPS framework uses cloud foundations to give cost viability, rapid organization, versatility, higher security, and adaptability to contain fluctuating workloads. This SMCPS system consists of two major steps: (i) Hybrid Key Generation (HKG) and (ii) storage management.

In order to increase the security of the cloud, the initial step of the work encrypt the cloud medical healthcare records by using the Hybrid Key Generation (HKG) algorithm, and the encrypted key values are stored in the data owner phase. This proposed Hybrid Key Generation (HKG) algorithm is different from the usual key generation algorithm, since the proposed HKG algorithm adds the key values of AES and MD5. A new key value is created from these two algorithms, thus increasing the security level of the system. In the second step of the work, a Compressive Sensing (CS) based framework for storage management of medical healthcare records was developed. The distributed matching engine attains more scalability as well, and it is developed to among diverse healthcare records that are being collected from the UCI machine learning repository. Experimentation work was performed by using the Signal and Image Processing Institute (SIPI) database for the watermark pattern, and any medical healthcare records were taken as input in the MCC environment. Results demonstrated that the safe watermark recognition in the CS domain is possible for medical healthcare records.

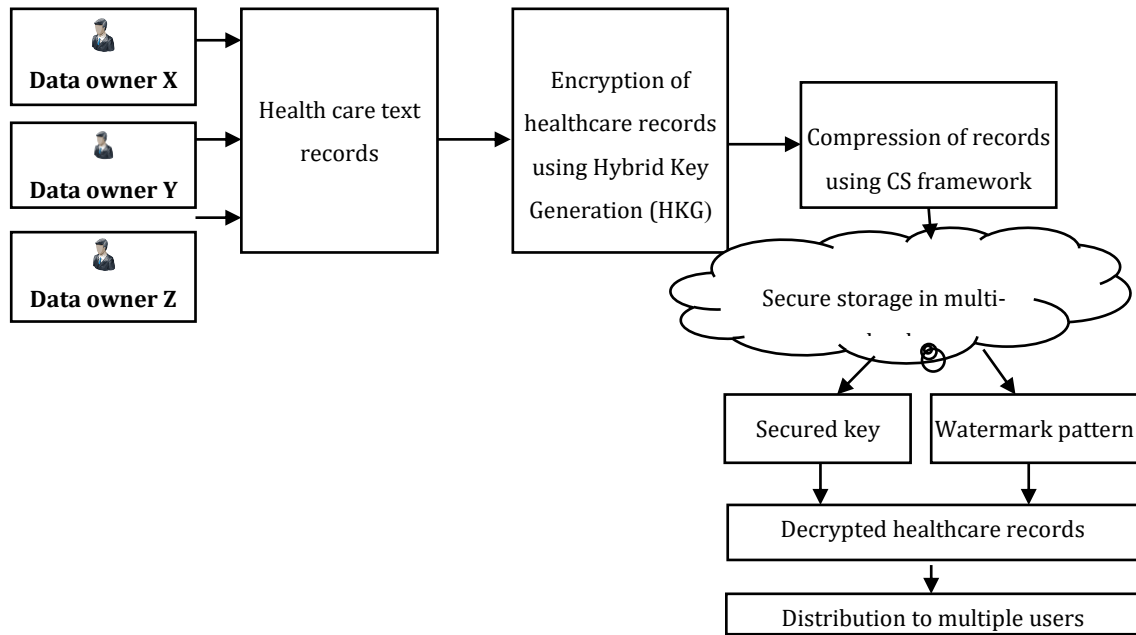


Figure 1: Secure Medical Healthcare Content Protection System (SMCPS)

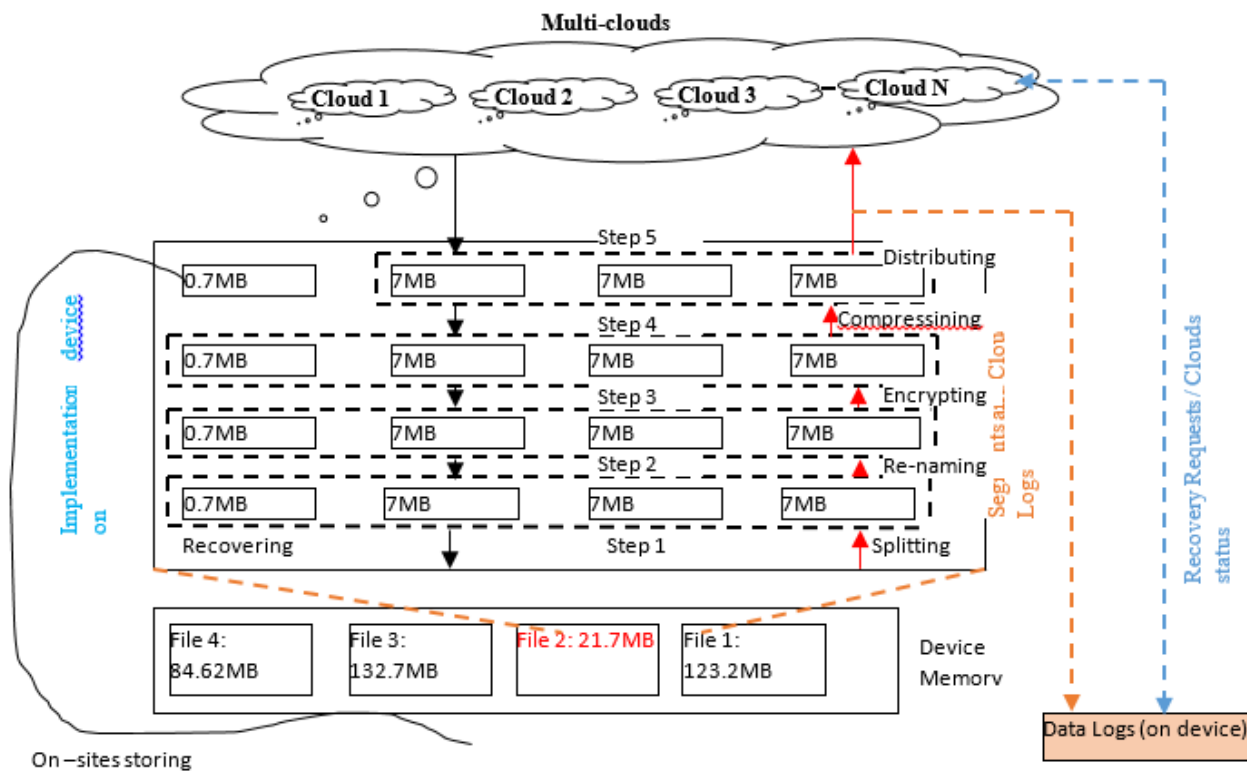


Figure 2: The workflow of operations

Figure 2 demonstrates the workflow of operations. It illustrates how the medical healthcare records could interact and keep informed of the multi-cloud status; and at wherein stage the healthcare records on-site storage and healthcare records need to be accomplished.

3.1 Multi-clouds model

As stated above, this presented SMCPs technique is based on the multi-clouds conception; with the intention that, there are an amount of a variety of free clouds would be used to construct the multi-clouds model. According to the service-based model, the developers would develop a service, which would work more similarly to a third party among the users and the clouds, and that could be illustrated in Figure 3. Here, the developed service (mediator) allocates resources among the user and the clouds.

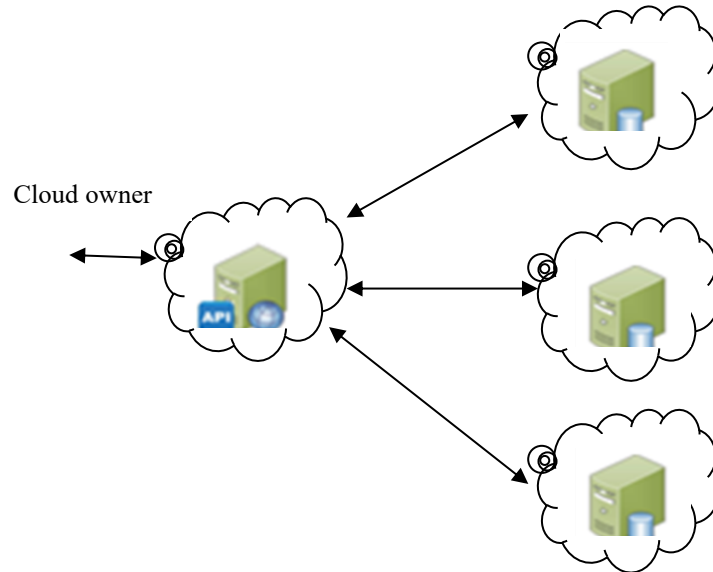


Figure 3: Multi-cloud service-based architecture

3.2 Data Splitting

In the data splitting, we divide the healthcare data file into segments is most significant for various reasons:

- i. Sharing out over the multi-clouds
- ii. Evadecloud maximum-size file issue

Additionally, dividing the healthcare records would improve the use of the offered bandwidth and enable load balancing. In addition, it would enable the encrypt/decryption processes by decreasing the size of the computed files. Officially, the health care records division enhances the processing throughput, by letting the pipeline and parallel processes that would decrease the operational cost as well as the consumed time.

3.3 Encrypt/Decrypt

To enhance the offered security and assure the integrity of the healthcare records, the presented method would consist of encryption methods. Utilizing Hybrid Key Generation (HKG) would improve the protection of the developed API level. As a consequence of diversity in medical healthcare records abilities and the user's security needs, there are a number of cryptography techniques that would help to facilitate the user selecting the appropriate method, and key length as well. HKG is introduced in this work, which combines the procedure of data encryption methods such as Advanced Encryption Standard (AES) and MD5 for encrypting healthcare records [15].

3.3.1 Advanced Encryption Standard (AES)

Advanced Encryption Standard, simply AES, is used to enhance encryption performance. On the other hand, AES simply lets a 128-bit healthcare record length, which could be split into four fundamental operation blocks. Such blocks work on an array of bytes and are prearranged as a 4×4 matrix, which is known be the state. For complete encryption, data has been sent through Nr rounds as (Nr = 10, 12, 14). These rounds have been directed by means of subsequent transformations:

(i) Bytesub transformation: It is a type of non-linear byte substitution, using a substitution table, namely s-box, which has been built via multiplicative inverse and the affine transformation.

(ii) Shiftrows transformation: It is one of the easy byte transpositions, bytes in the final three rows of the state are as clockwork shift; the offset of left shift could change from one to three bytes.

(iii) Mixcolumns transformation: It is similar to matrix multiplication of the columns of states. Each column vector has been multiplied by a preset matrix. It should be noted that bytes are treated as polynomials that are willingly than the numbers. The transformation could be determined by following matrix multiplication on a state.

$$\begin{bmatrix} 02 & 03 & 01 & 01 \\ 01 & 02 & 03 & 01 \\ 01 & 01 & 02 & 03 \\ 03 & 01 & 01 & 02 \end{bmatrix} \begin{bmatrix} s_{0,0} & s_{0,1} & s_{0,2} & s_{0,3} \\ s_{1,0} & s_{1,1} & s_{1,2} & s_{1,3} \\ s_{2,0} & s_{2,1} & s_{2,2} & s_{2,3} \\ s_{3,0} & s_{3,1} & s_{3,2} & s_{3,3} \end{bmatrix} = \begin{bmatrix} s'_{0,0} & s'_{0,1} & s'_{0,2} & s'_{0,3} \\ s'_{1,0} & s'_{1,1} & s'_{1,2} & s'_{1,3} \\ s'_{2,0} & s'_{2,1} & s'_{2,2} & s'_{2,3} \\ s'_{3,0} & s'_{3,1} & s'_{3,2} & s'_{3,3} \end{bmatrix} \quad (1)$$

$$s'_{0,j} = (2 \cdot s_{0,j}) \oplus (3 \cdot s_{1,j}) \oplus s_{2,j} \oplus s_{3,j} \quad (2)$$

$$s'_{1,j} = s_{0,j} \oplus (2 \cdot s_{1,j}) \oplus (3 \cdot s_{2,j}) \oplus s_{3,j} \quad (3)$$

$$s'_{2,j} = s_{0,j} \oplus s_{1,j} \oplus (2 \cdot s_{2,j}) \oplus (3 \cdot s_{3,j}) \quad (4)$$

$$s'_{3,j} = (3 \cdot s_{0,j}) \oplus s_{1,j} \oplus s_{2,j} \oplus (2 \cdot s_{3,j}) \quad (5)$$

Where • represents multiplication over a finite field. If the label and these, namely A and the A-1, in respectively labeled and label state, which is before the mix columns operation, such as S, and also after S'.

$$AS = S' \quad (6)$$

Therefore

$$A^{-1}S' = A^{-1}AS = S \quad (7)$$

(iv) Addroundkey transformation: It is an easy XOR between the working state and the roundkey.

Algorithm 1. Key expansion (byte Key(16), word w(44))

```
{
  Word Temp
  For(i = 0; i < 4; i++)
    For(j = 0; j < 44; j++)
      w[j] = (key[4 * i], key[4 * i + 1], key[4 * i + 2], key[4 * i + 3]);
}
```

```

Temp=w[i-1];
If(i mod 4=0)

Temp=Sub word (Rotword(temp))  $\oplus R \left[ \frac{i}{4} \right]$ ;

w[i]=w[1-4]  $\oplus$  temp;
}
}

```

The key has been copied as the first four words of an expanded key. The remainder of the expanded key has been filled in four words at a time. Each add word $w[i]$ could depend on an immediately preceding word, namely $w[i - 1]$, and word four positions back as $w[i - 4]$. In four cases, three cases are using simple XOR. For a word where the position in the w array is a multiple of 4, it is a more complex function.

3.3.2 MD5 message-digest

MD5 could process variable-length healthcare records in a fixed-length output of 128 bits. The input healthcare record has been broken up into chunks of 512-bit blocks, and the healthcare record has been padded so that its own length is divisible by 512. The first single bit, namely 1, has been appended to the end of healthcare records. This has been followed by many zeros that are required for bringing the length of healthcare records up to 64 bits, which is less than a multiple of 512. The remaining bits have been filled along 64 bits, which denotes the length of an original healthcare record as a modulo 2^{64} .

3.3.3 HKG

The AES algorithm is used to provide security of healthcare data through encryption and the decryption process. The MD5 helps generate an additional secret key for the data owner and the cloud user for communication.

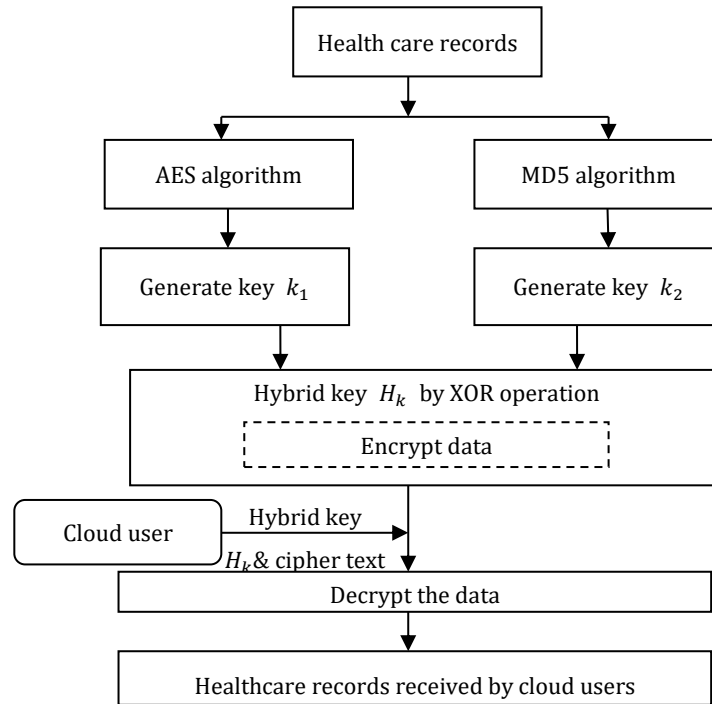


Figure 4: HKG algorithm architecture diagram

In this proposed HKG algorithm, by using bit-wise XOR operation, the key values of AES and MD5 algorithms are combined, and a new key value is created. This XOR operation increases the security of the cloud user's healthcare records. This operation is performed by means of after healthcare records are converted into ciphertext. In this proposed approach, first choose healthcare records, then key values k_1 & k_2 are created from MD5, and AES. This would be used to perform the XOR operation. Then, finding out the encryption and a decryption key could exponent and that has been used for an encryption and also a decryption process. At the sender side, an encryption is being done by an encryption algorithm. When encryption starts, the XOR operation might be performed between the k_1 & k_2 . A new key value has been created and then named H_k . After the operation secret message has been sent via a medium. At the receiver side secret key and secret message that given by the cloud user to receive the original health care records. Then decrypt the original file using the hybrid key, ciphertext text and get the original healthcare records sent by the sender. The diagrammatic representation of the proposed HKG algorithm is shown in the figure.

3.4 CS transformation

There are mainly three parties in the presented structure, the Data Holders, simply DH of the possibly called watermarked images, Watermark Owners called WO, and Cloud, namely CLD, as depicted in Figure 5. The structure as well needs a Certificate Authority, and simply as CA for bringing out public keys as well as CS matrix keys to assure named parties of the structure. For DH, namely media agencies, while it gathers a huge amount of medical healthcare data from the Internet and also stores an encrypted version in CLD, it also needs to ensure that those medical healthcare records can be edited and also republished officially. Watermark Owners, or shortly as WOs, provide that providers who can share the watermarked content, which means watermark embedding has been carried out by means of WO, previous to the content being published. WOs at all times desire to make out if their contents are lawfully utilized and republished.

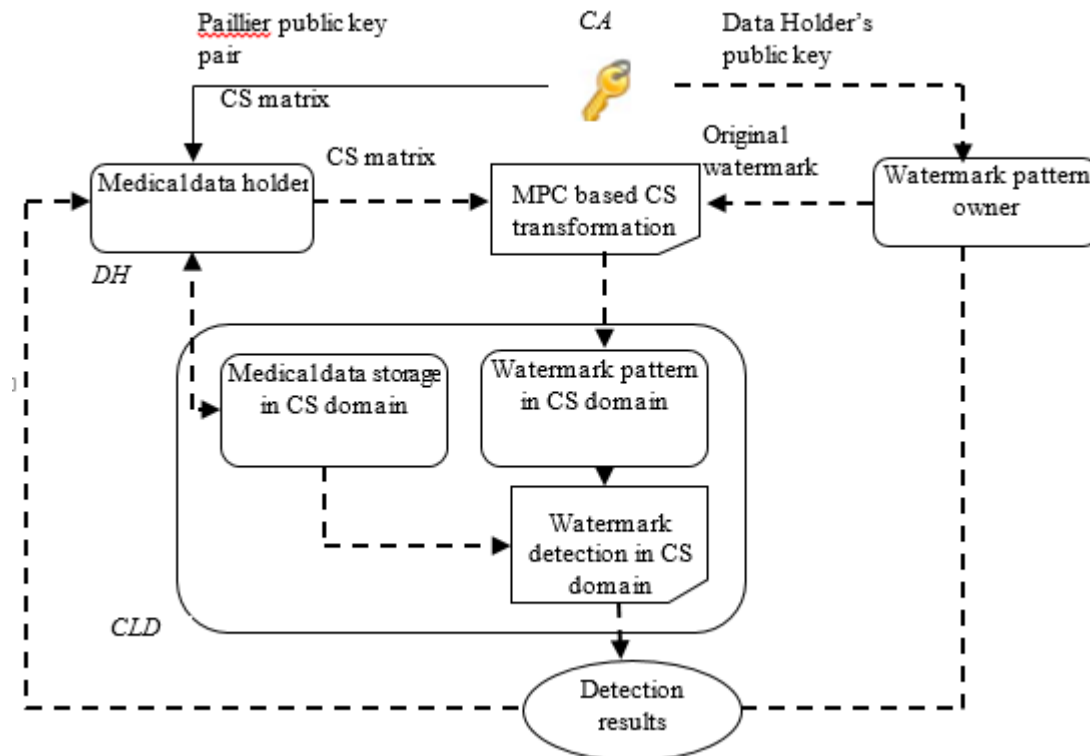


Figure 5: Architecture of the proposed CS framework

Primarily, the CA wants to bring out the CS matrix matches to DH. The CS matrix also comprises seeds and a random function utilized to produce the Gaussian CS matrix. Utilize CA to bring out a random function to ensure the randomness of the produced Gaussian CS matrix. The CA, as well, wants to bring out an AES and MD5 pair to DH and DH's public key for WO. Generally, DH as well contains a diverse private compressive sensing matrix (derived from the seed) for every image. DH changes the image DCT coefficients to the compressive sensing domain, as well as considers CLD that contains CS domain data for storage. In case watermark recognition with the WO is needed, we want CLD to contain the watermark in an identical CS domain that has been attained by running a safe multiparty protocol.

Compressive Sensing (CS) Framework: The Restricted Isometry Property, called RIP, is a needed condition particularly for perfect reconstruction in compressive sensing theory. Before presenting compressive sensing theory, first of all, RIP has to be introduced.

Restricted Isometry Property (RIP): A vector called $x \in R^n$ is a k-sparse and if $|\{j : |x_j| > 0\}| \leq k$. A matrix called $\Phi \in R^{m \times n}$ is called having RIP of an order k and the level $\delta \in (0, 1)$ (which is equivalently, and (k, δ)-RIP) if,

$$(1 - \delta) \|x\|_2^2 \leq \|\Phi x\|_2^2 \leq (1 + \delta) \|x\|_2^2 \quad (8)$$

For all the k-sparse $x \in R^n$. The restricted isometry constant called δ_k has been defined as the smallest value of δ for above inequality holds.

The CS theory [16] asserts that while data can be represented using a small number of the nonzero coefficients, they can be perfectly recovered after being transformed by a limited number of incoherent, non-adaptive linear measurements. Suppose, data called $d \in R^n$ is a k-sparse vector, which means that only k elements are out of n elements of the data nonzero element, and transformed for $d \in R^m$, whereas $m < n$ $x = \Phi_{m \times n} d$. If $\Phi_{m \times n}$ might satisfy RIP, it has been shown by solving the optimization problem:

$$\min \|d\|_1 \text{ s.t. } x = \Phi_{m \times n} d \quad (9)$$

Equation (9) is found as equivalent for finding out the sparsest solution called d to $x = \Phi_{m \times n} d$ providing $m \geq Ck \log(n/k)$, whereas C is small constant. Eq. (9) can present an L1 minimization problem that can be solved through Orthogonal Matching Pursuit and simply OMP algorithms. It is also shown, there are such many ways to construct a matrix, namely $\Phi_{m \times n}$ Which meets RIP property, for instance, if entries of the matrix called $\Phi_{m \times n}$. Those have been generated from a Gaussian distribution with a zero mean and a variance such as $1/m$, $\Phi_{m \times n}$ is RIP matrix along an overwhelming probability. In this particular framework, the CS matrix was generated from a Gaussian distribution. Let, the $X = \{x_i\}$, $Y = \{y_i\}$ be a two n-dimensional cloud user data vectors and the $\Phi_{m \times n}$ be an $m \times n$ -dimensional random matrix. Each particular entry of a $\Phi_{m \times n}$ is found as independent and identically chosen from the Gaussian distribution with the mean zero and the variance $1/m$. In further, the $P = \Phi_{m \times n} X$; and the $R = \Phi_{m \times n} Y$. Then,

$$E[X^T Y - P^T R] = 0 \quad (10)$$

$$\text{Var}[X^T Y - P^T R] = \frac{1}{m} (\sum_{i=1}^n x_i^2 \sum_{i=1}^n y_i^2 + (\sum_{i=1}^n x_i y_i)^2) \quad (11)$$

In Zeng and Liu's work [17], the most excellent watermark recognition performance has been attained by means of utilizing the selective DCT coefficients for the recognition. On the other hand, DH in the framework may not have information that are regarding the criteria for the selection procedure. Still, if we are presuming that DH could contain such type of information, DH can want to know DCT channels that are being used for the watermark recognition. Embedding process as $d'_i = d_i + w_i$, whereas

w_i is an inserted watermark signal that is derived from the watermark pattern $\{y_i\}$. In the detection process, test cloud user data $\{x_i\}$ has correlated along with the watermark pattern $\{y_i\}$. Detection of watermarks has been accomplished through hypothesis testing.

$H_0: x_i = d_i + n_i$ without watermark

$H_0: x_i = d_i + n_i + w_i$ with watermark

Where n_i is noise. Normalized correlating detector can output a test statistic called q , which has been compared with a threshold called T for identification, if the test image could contain claimed watermarks:

$$q = \frac{\sum_{i=1}^n z_i}{V_{z_i} \sqrt{n}} = \frac{M_{z_i} \sqrt{n}}{V_{z_i}} \quad (12)$$

Where as $z_i = x_i y_i$, namely n is the size of the feature set called $\{z_i\}$. M_{z_i} and the $V_{z_i}^2$ are the sample mean and the sample variance of term z_i that given as follows:

$$M_{z_i} = \frac{\sum_{i=1}^n z_i}{n}, V_{z_i}^2 = \frac{\sum_{i=1}^n (z_i - M_{z_i})^2}{n-1} \quad (13)$$

Finally, under the hypothesis called H_0 , for the large n , q has approximately a normal distribution, such as $q \sim N(0, 1)$. Under the H_1 , $q \sim N(\mu, 1)$, whereas $\mu > 0$.

4 Experimental Results

In the experimental work assessed the presented SMCPs system was assessed utilizing some standard images, such as Peppers, Baboon, and Lena, with 512×512 images. These standard watermark pattern images were collected from the standard benchmark Signal and Image Processing Institute (SIPI) database [18]. The SIPI database comprises 44 images, 16 colors, and a total of 28 monochromes. The sizes are 14 256×256 , 26 512×512 , and 4 1024×1024 . For an individual image, the first click on an image name, which is left of every line below for shows a 200×200 pixel preview of the image. The 24-bit color-based images have been reduced to 8 bits/pixel in order to previewing purposes. The proposed algorithms are implemented by using a JAVA environment with an SQL database and an implemented Glowfish server.

Baboon: Courtesy of Signal and the Image Processing Institute at the University of Southern California is illustrated in Figure 6.



Figure 6: Baboon Samples

Lena: Courtesy of a Signal and an Image Processing Institute at the University of Southern California, and shown in Figure 7.



Figure 7: Lena input image

Peppers: Courtesy of the Signal and Image Processing Institute at the University of Southern California is shown in Figure 8.



Figure 8: Pepper's input image

Select the one wherein the watermark pattern used for the watermark recognition is openly produced, particularly from a Normal distribution such as $N(0, 1)$. The CS matrix size is exceedingly huge, only if that transfers a 512×512 image for the vector to CS transformation. To replace specifically for cutting an image into pieces, and each piece possesses $64 \times 8 \times 8$ DCT blocks.

Patient Name	poiuyt
Profession	Student
Age	26
Address	ssss
Mobile Number	9876545612
Email Id	dselvait@gmail.com
Disease	ddetbvf
Solution	sdfh
Status	gdhxfghgh

Figure 9: Patient attributes information

The Selective DCT coefficients of every piece could form a vector and also as changed to the CS domain at a similar as CS rate in however, it utilizes the diversity of CS matrices. In the watermark pattern, the healthcare dataset samples were collected from the benchmark UCI machine learning repository. Many healthcare patients' records were collected from this repository. In addition, the research work on healthcare patients' records is encrypted by adding attributes of the patients. The sample attributes of patients are shown in Figure 9. In order to secure patient attributes and the patient healthcare records; the files are encrypted using HKG and shown in Figure 10.

Patient Name	[B@1ec344e
Profession	[B@5e0dab
Age	[B@d36c4e
Address	[B@6b9f5d
Mobile Number	[B@1abae94
Email Id	[B@52a176
Disease	[B@e660b5
Solution	[B@d73ed3
Status	[B@15db81b

Figure 10: Patient attributes information with key values



Figure 11: Watermark patterns

In order to provide more security from unauthorized users, the proposed model encrypts records and adds watermarks by using the standard watermark images, are shown in Figure 11. Once the watermark patterns are selected, the files are embedded in the image, and it is shown in Figure 12.

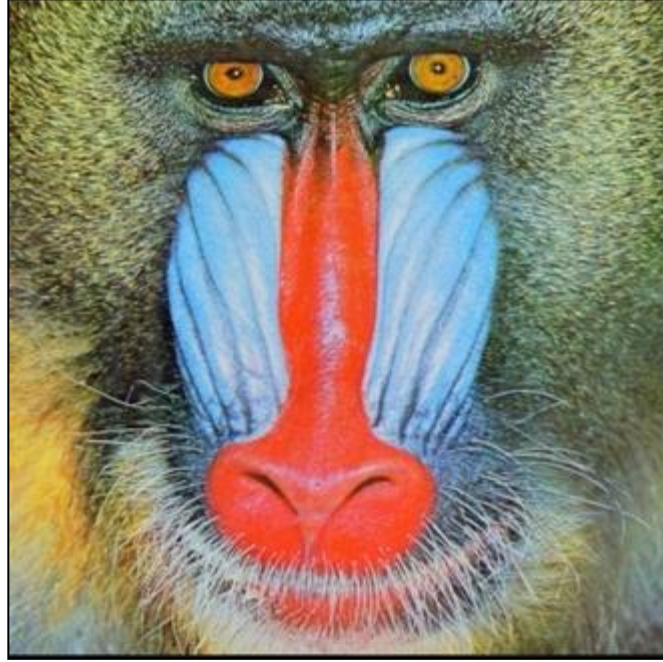


Figure 12: Watermark image with the healthcare data

4.1 Compression methods comparison

Macro-scale compression, zero-knowledge watermark protocol, CS framework, and the proposed SMCPS framework are compared in this evaluation. CS based privacy preserving framework, as the DCT coefficients aren't entirely sparse, in which CS restoration would produce distortion for the reformed image, particularly when while CS rate is small. CS reformation error has been studied in a lot of particular works. Currently, MCPS with CS reformation experimentation outcomes, while each specific AC component has been modified to the CS domain. The results are measured in terms of Peak Signal to Noise Ratio, which is shortly represented as PSNR, the Mean Squared Error, simply MSE, and the Structural Similarity Index called SSIM.

4.2 Mean Squared Error (MSE)

PSNR is as most in easily defined through Mean Squared Error, called MSE. Given a noise-free $m \times n$ monochrome image such as I and that noisy approximation called K , MSE is also defined by

$$MSE = \frac{1}{mn} \sum_{i=0}^{m-1} \sum_{j=0}^{n-1} [I(i, j) - K(i, j)]^2 \quad (14)$$

4.3 Peak Signal to Noise Ratio(PSNR)

Peak Signal to Noise Ratio is found as the most commonly used measure quality of the reconstruction of lossy compression codecs. The signal is original data, and the noise is an error introduced by means of compression. While comparing compression codecs, the PSNR is an approximation for human perception of the reconstruction quality. Although by higher PSNR generally indicates reconstruction is found as higher in quality, in some cases it might not. One has to be extremely careful with the range of validity of this particular metric; it is only conclusively valid while it is utilized for comparing results from the same codec and the same content. The PSNR that represented in dB, has been defined as:

$$PSNR = 10 \log_{10} \left(\frac{MAX_I^2}{MSE} \right) = 20 \log_{10} \left(\frac{MAX_I}{\sqrt{MSE}} \right) \quad (15)$$

$$PSNR = 10 \log_{10}(MAX_I) - 10 \log_{10}(MSE) \quad (16)$$

4.4 Structural Similarity (SSIM) index

SSIM is for measuring similarity between the two images. SSIM index is a full reference metric; namely, in other words, the measurement or the prediction of an image quality is purely based on an initial uncompressed or a distortion-free image as by reference. The SSIM index has been calculated on the various windows of the image. The measurement between two windows, called x and y , in a common size called $N \times N$, is

$$SSIM(x, y) = \frac{(2\mu_x\mu_y + c_1)(2\sigma_{xy} + c_2)}{(\mu_x^2 + \mu_y^2 + c_1)(\sigma_x^2 + \sigma_y^2 + c_2)} \quad (17)$$

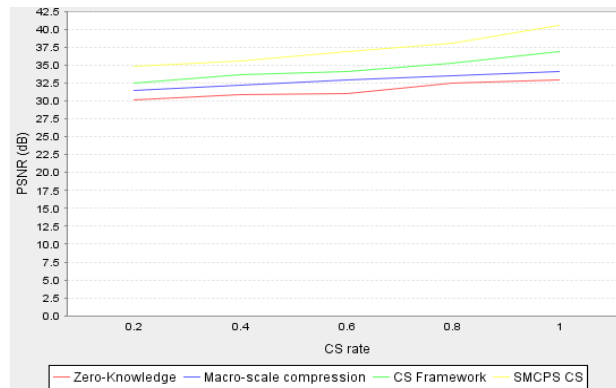


Figure 13: Comparison of the PSNR vs. other algorithms

The experimentation results prove that the Peak Signal-to-Noise Ratio of the proposed SMCPS framework is about 40.5 dB, whereas the other compression methods, such as Macro-scale compression, zero-knowledge watermark protocol, and CS framework, produce PSNR results of 32.89 dB, 34.15 dB, and 36.95 dB, while the CS rate is 1.0, as depicted in Figure 13 (See table 2). It concludes the proposed SMCPS framework, which performs better when compared to the other compression methods.

Table 1: PSNR vs. other algorithms

CS rate	PSNR(dB)			
	Zero-knowledge	Macro-scale compression	CS Framework	SMCPS-CS
0.2	30.12	31.52	32.5	34.8
0.4	30.87	32.15	33.69	35.51
0.6	31.05	32.98	34.1	36.89
0.8	32.46	33.5	35.25	38
1	32.89	34.15	36.95	40.5

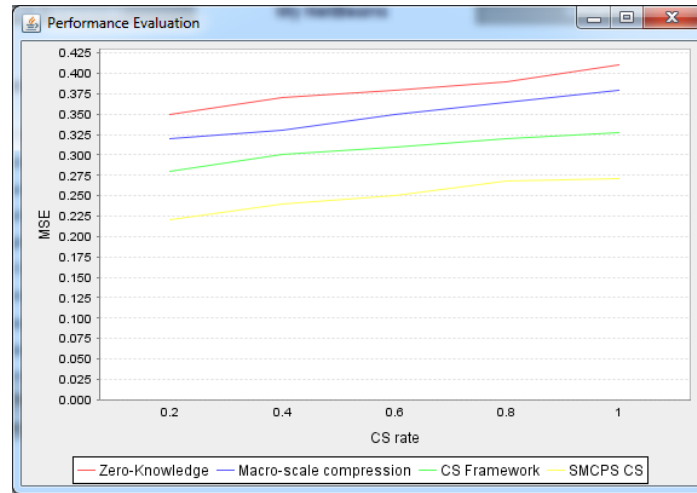


Figure 14: MSE Comparison vs. other algorithms

Figure 14 shows the MSE performance comparison results of Macro-scale compression, zero-knowledge watermark protocol, CS framework, and proposed SMCPs framework. However, the proposed SMCPs framework produces MSE results of 0.271, whereas the other compression methods, such as Macro-scale compression, zero-knowledge watermark protocol, and CS framework, produce MSE results of 0.41, 0.38 dB, and 0.328, at a CS rate is 1.0 (See table 2). It concludes that the proposed SMCPs framework produces a lower MSE value when compared to other compression methods.

Table 2: MSE vs. other algorithms

CS rate	MSE			
	Zero-knowledge	Macro-scale compression	CS Framework	SMCPs-CS
0.2	0.35	0.32	0.28	0.22
0.4	0.37	0.33	0.3	0.24
0.6	0.38	0.35	0.31	0.25
0.8	0.389	0.365	0.32	0.268
1	0.41	0.38	0.328	0.271

Figure 15 shows the SSIM performance comparison results of Macro-scale compression, zero-knowledge watermark protocol, CS framework, and proposed SMCPs framework. From the results, the proposed SMCPs framework produces SSIM results of 0.8581, whereas the other compression methods, such as Macro-scale compression, zero-knowledge watermark protocol, and CS framework, produce MSE results of 0.76, 0.7825, and 0.823, at a CS rate is 1.0 (See table 3). It concludes that the proposed SMCPs framework produces a lower MSE value when compared to other compression methods.

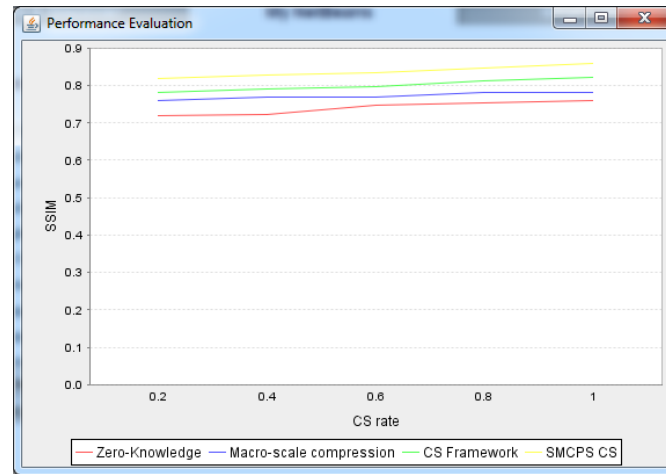


Figure 15: SSIM Comparison vs. other algorithms

Table 3: SSIM vs. other algorithms

CS rate	SSIM			
	Zero-knowledge	Macro-scale compression	CS Framework	SMCPs-CS
0.2	0.72	0.76	0.78	0.82
0.4	0.723	0.768	0.791	0.828
0.6	0.746	0.77	0.798	0.834
0.8	0.753	0.78	0.814	0.8478
1	0.76	0.7825	0.823	0.8581

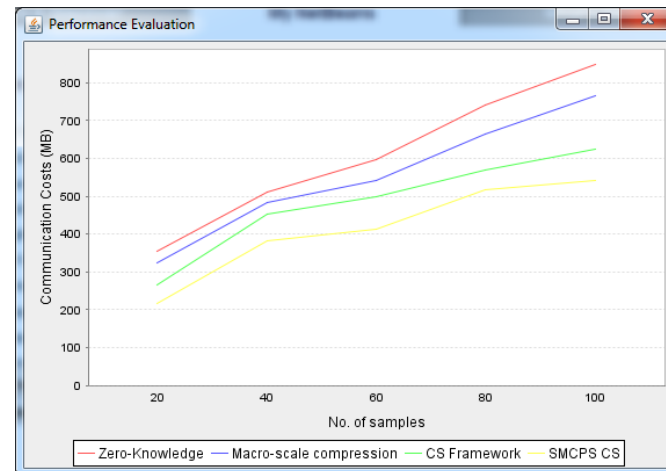


Figure 16: Comparison of the communication costs vs. other algorithms

In Figure 16 prove that as the amount of healthcare samples for safe watermark recognition increases, the proposed SMCPs is compared with existing methods in terms of communication cost (See table 4). SMCPs structure contains much superior scalability as well as superior competence while carrying out safe watermark identification on a huge amount of healthcare samples, although the SMCPs framework is superior to other algorithms.

Table 4: Communication costs vs. other algorithms

No. of samples	Communication costs(MB)			
	Zero-knowledge	Macro-scale compression	CS Framework	SMCPS-CS
20	356	325	265	218
40	512	484	452	381
60	598	541	498	412
80	742	665	569	518
100	847	765	625	543

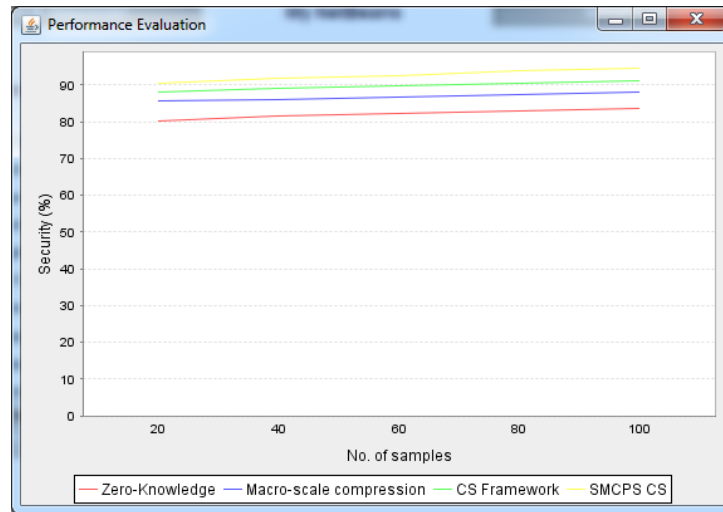
**Figure 17:** Comparison of the security level vs. other algorithms

Figure 17 proves that as the amount of healthcare samples for safe watermark detection enhances, the presented Secure Multimedia Content Protection System (SMCPS), existing methods in words of security (See table 5). The SMCPS framework provides higher security while carrying out safe watermark identification on a large number of healthcare samples.

Table 5: Security vs. other algorithms

No. of samples	Security(%)			
	Zero-knowledge	Macro-scale compression	CS Framework	SMCPS-CS
20	80.26	85.69	88.13	90.58
40	81.58	86.15	89.18	91.81
60	82.25	86.54	89.93	92.57
80	83.02	87.41	90.42	93.78
100	83.58	88.17	91.23	94.46

5 Conclusion

In this review, a new Secure Medical Healthcare Content Protection System (SMCPS) is proposed for a Multi-Cloud Computing (MCC) environment. This SMCPS uses two different

encryption algorithms, such as AES and MD5 encryption, which create a new key by multiplying key 1 and key 2, which increases the security and efficiency. The proposed SMCPs system possesses two major steps, namely, at an initial step, the key values are created for healthcare records by encryption algorithms for secure computation, and the second step is a Compressive Sensing (CS) based framework that is proposed for data storage management in the MCC environment. CS framework reduces the size of the original medical or healthcare content for image acquisition. From the results, it is concluded that the proposed SMCPs framework produces MSE results of 0.271, whereas the other compression methods, such as Macro-scale compression, zero-knowledge watermark protocol, and CS framework, produce MSE results of 0.41, 0.38 dB, and 0.328, at a CS rate is 1.0. In future work, the proposed SMCPs system work is to be enhanced by including other encryption methods, which might assist in increasing the security level. Finally, the work intends to expand the MCC to the proposed SMCPs system to their individual cloud storage.

Acknowledgement: Not applicable.

Funding Statement: Not applicable.

Author Contributions: The authors confirm contribution to the paper as follows: Conceptualization, methodology and writing—original draft preparation: S. Sivakumar; supervision and writing—review and editing: T. Karthikeyan. All authors reviewed the results and approved the final version of the manuscript.

Conflicts of Interest: The authors declare no conflicts of interest to report regarding the present study.

References

1. Senyo PK, Addae E, Boateng R. Cloud computing research: A review of research themes, frameworks, methods, and future research directions. *Int J Inf Manag*. 2018;38(1):128-39.
2. Buyya R, Srirama SN, Casale G, Calheiros R, Simmhan Y, Varghese B, Gelenbe E, Javadi B, Vaquero LM, Netto MA, Toosi AN. A manifesto for future generation cloud computing: Research directions for the next decade. *ACM Comput Surv*. 2018;51(5):1-38.
3. Ullah A, Nawi NM, Ouham S. Recent advancement in VM task allocation system for cloud computing: review from 2015 to 2021. *Artif Intell Rev*. 2022;55(3):2529-73.
4. Suryateja PS. Threats and vulnerabilities of cloud computing: a review. *Int J Comput Sci Eng*. 2018;6(3):297-302.
5. Dawood M, Tu S, Xiao C, Alasmay H, Waqas M, Rehman SU. Cyberattacks and security of cloud computing: a complete guideline. *Symmetry*. 2023;15(11):1981.
6. Mohammad HM, Abdullah AA. Enhancement process of AES: a lightweight cryptography algorithm-AES for constrained devices. *TELKOMNIKA*. 2022;20(3):551-60.
7. Kasgar AK, Dhariwal MK, Tantubay N, Malviya H. A review paper of message digest 5 (MD5). *Int J Mod Eng Manag Res*. 2013;1(4):29-35.
8. Kim D, Kwon H, Hahn C, Hur J. Privacy-preserving public auditing for educational multimedia data in cloud computing. *Multimed Tools Appl*. 2016;75(21):13077-91.
9. Qin Z, Yan J, Ren K, Chen CW, Wang C. SecSIFT: Secure image SIFT feature extraction in cloud computing. *ACM Trans Multimed Comput Commun Appl*. 2016;12(4s):1-24.
10. Voloshynovskiy S, Diephuis M, Holtyak T. Privacy preserving multimedia content identification for cloud-based bag-of-feature architectures. In 2015 IEEE International Conference on Multimedia & Expo Workshops (ICMEW) 2015 Jun 29, Turin, Italy, 2015, p. 1-6.
11. Zhang Q, Yang LT, Chen Z. Privacy preserving deep computation model on cloud for big data feature learning. *IEEE Trans Comput*. 2015;65(5):1351-62.
12. Kang LW, Muchtar K, Wei JD, Lin CY, Chen DY, Yeh CH. Privacy-preserving multimedia cloud computing via compressive sensing and sparse representation. In 2012 International Conference on Information Security and Intelligent Control 2012 Aug 14, Yunlin, Taiwan, 2012, p. 246-249.

13. Alqahtani HS, Sant P. A multi-cloud approach for secure data storage on smart device. In 2016 sixth international conference on digital information and communication technology and its applications (DICTAP) 2016 Jul 21, Konya, Turkey, 2016, p. 63-69.
14. Liu X, Choo KK, Deng RH, Yang Y, Zhang Y. PUSC: Privacy-preserving user-centric skyline computation over multiple encrypted domains. In 2018 17th IEEE International conference on trust, security and privacy in computing and communications/12th IEEE International conference on big data science and engineering (TrustCom/BigDataSE) 2018 Aug 1, New York, NY, USA, 2018, p. 958-963.
15. Kumar R, Mahajan G. A novel framework for secure file transmission using modified AES and MD5 algorithms. *Int J Inf Comput Secur.* 2015;7(2-4):91-112.
16. Qaisar S, Bilal RM, Iqbal W, Naureen M, Lee S. Compressive sensing: From theory to applications, a survey. *J Commn Net.* 2013;15(5):443-56.
17. Zeng W, Liu B. A statistical watermark detection technique without using original images for resolving rightful ownerships of digital images. *IEEE Trans Image Process.* 1999;8(11):1534-48.
18. Yousif SF, Abboud AJ, Radhi HY. Robust image encryption with scanning technology, the El-Gamal algorithm and chaos theory. *IEEE Access.* 2020;8:155184-209.